

# Hyperlexia in Children and Adolescents: A Comprehensive Narrative Review of Definitions, Prevalence, Mechanisms, Comprehension, Allied Strengths, and Strength-Based Intervention

*With design implications for a bilingual (English / Québec French), music-integrated game suite for an adolescent reader*

Research compilation prepared with Claude (Anthropic AI research assistant) for family and educator use

Compiled 27 September 2026    120 sources    86 hyperlexia records    15 figures    8 tables    Languages of evidence: EN, FR, ES, PT, IT, JA, ZH

## ABSTRACT

**Background.** Hyperlexia — precocious, largely self-taught word reading that outstrips comprehension, together with an intense orientation toward written material — was first described in 1967 and is still debated as a disability, a "superability", or a learning style. This review compiles every unique finding available across the indexed literature into a single reference.

**Methods.** Narrative synthesis of all Europe PMC records mentioning *hyperlexia* in their title or abstract (86 unique records, 1967–2025, drawn from 236 full-text hits), supplemented by 24 targeted studies on reading comprehension in autism, figurative language, bilingualism, music, and twice-exceptional adolescents, 2 foundational framework papers, and 8 clinical/educational sources (120 sources in total). Quantitative results were extracted verbatim from the primary reports and re-plotted; no new data were collected and no pooling was attempted.

**Results.** Hyperlexia is reported in about 6–21% of autistic children, and 84% of published hyperlexic case reports involve autism, yet it also appears in typical development, ADHD, Turner and 22q11.2 deletion syndromes, Prader-Willi, West syndrome, tuberous sclerosis, and — as a distinct acquired sign — after frontal lesions in adults. In a Montréal clinic cohort, interest in letters emerged at a median of 30 months in autistic children (the same age as in typical development) even though 76% were minimally verbal; the odds of stronger interest were 2.78× for letters and 3.49× for numbers relative to non-autistic clinical peers. Mechanistic work converges on an orthographic, perception-first route to word recognition that can function without semantic access, with fMRI showing hyperactivation of left superior temporal cortex. Comprehension difficulty tracks semantic/oral-language knowledge (57% of variance) and is greatest for socially loaded, inferential and figurative text. Interventions that lead with print — written prompts, word–picture matching, question scaffolding, graphic representations ( $\text{Tau-U} = 0.85$ ) — show consistent gains. Numbers (hypercalculia profile in 20% of one national sample), music (absolute pitch in 5–11% of autistic people; music-therapy global improvement  $\text{RR} = 1.22$ ) and additional languages are allied strengths.

**Conclusions.** Hyperlexia is best treated as a heterogeneous, strength-anchored reading profile. Effective support "writes it out", teaches meaning, inference and pragmatics explicitly, and harnesses allied systemising interests. We translate these findings into nine design principles and a 25-game bilingual suite built for a 15-year-old hyperlexic, hypernumerate pianist.

**Keywords:** hyperlexia; autism; reading comprehension; simple view of reading; orthographic processing; hypernumeracy; music; absolute pitch; bilingualism; Québec French; strength-based intervention; serious games

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## 1 INTRODUCTION

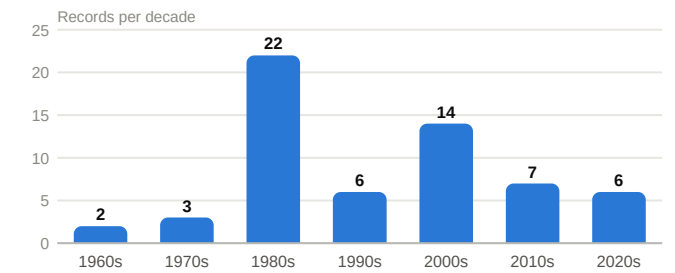
### 1.1 A short history

In 1967 Norman and Margaret Silberberg described young children whose word-recognition skill was far ahead of every other ability and named the phenomenon *hyperlexia* [1]. Within a year the field was already arguing about its meaning — Niensted asked whether it was "an educational disease" [2] — and early neurological series framed it as exceptional reading in children

with brain damage or developmental delay [3, 4]. Elliott and Needleman proposed a discrete "syndrome of hyperlexia" in 1976 [5].

The 1980s were the most productive decade (Fig. 1). Researchers reframed hyperlexia as a variant of developmental language disorder [6–8], dissected its reading routes [9–11], tested whether comprehension was really impaired [12], and ran the first population work in North Dakota [13, 14]. From 1999 onward, cognitive models [15, 16], neuroimaging [17] and systematic reviews [18, 19] followed. Much of the current evidence comes from two Montréal groups — Mottron, Ostrolenk and colleagues

[18, 20–22] and Macdonald, Luk and Quintin at McGill [23, 24] — which makes the literature unusually relevant to a French-speaking Québec family.



**Figure 1.** Records with *hyperlexia* in the title indexed in Europe PMC, by decade ( $n = 60$ ; the 2020s run to September 2026). The 1980s were the most active decade; recent output is concentrated in Montréal groups.

1.2 Why a single “mega” review

Findings about hyperlexia are scattered across neurology, psychiatry, linguistics, education and music cognition, and many papers are small case studies with one unique observation each. This review gathers every such observation in one place, re-plots the published numbers, and closes by translating the evidence into concrete design principles for a game suite commissioned for one reader: a 15-year-old hyperlexic adolescent who is bilingual in English and Québec French, reasons at roughly a 17-year-old level, counts into the nonillions, and plays the piano.

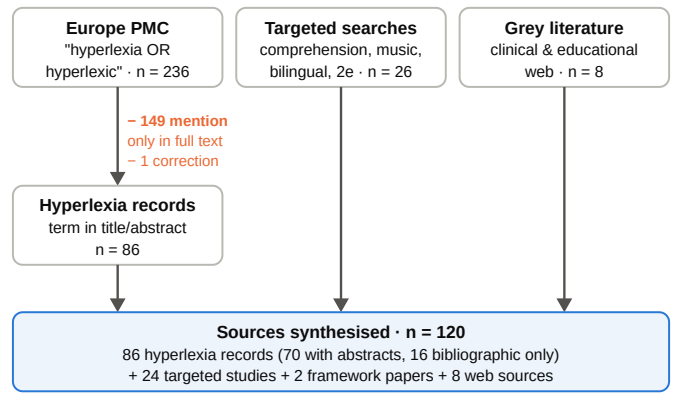
2 METHODS

2.1 Search strategy

The Europe PMC REST API was queried on 27 September 2026 for *hyperlexia* OR *hyperlexic* across all fields, returning 236 records. Records in which the term appeared only in full text (typically a passing mention) were set aside; the 87 records that used it in the title or abstract were retained, and one publisher correction was removed, leaving 86 unique hyperlexia records (60 with the term in the title). Targeted searches added the adjacent literatures that the hyperlexia papers themselves point to: reading comprehension in autism, inferencing and figurative language, bilingual language development, music perception and therapy, and twice-exceptional (2e) adolescents (24 records), plus two framework papers [25, 26]. Web searches identified eight clinical and educational sources that summarise current practice (Fig. 2).

2.2 Extraction and synthesis

For every record the design, sample, main quantitative results and — the focus of this review — its *unique* contribution were extracted (Appendix A). Sixteen older or non-English records had no indexed abstract; they are listed from their bibliographic data and flagged. Because definitions, measures and populations differ widely [19], results were synthesised narratively and numbers are re-plotted exactly as published, never pooled.



**Figure 2.** Source identification and selection. Unlike a formal systematic review, no records were excluded for quality; every hyperlexia record was retained and summarised (Appendix A).

3 DEFINITIONS AND CONCEPTUAL FRAMEWORKS

3.1 What counts as hyperlexia?

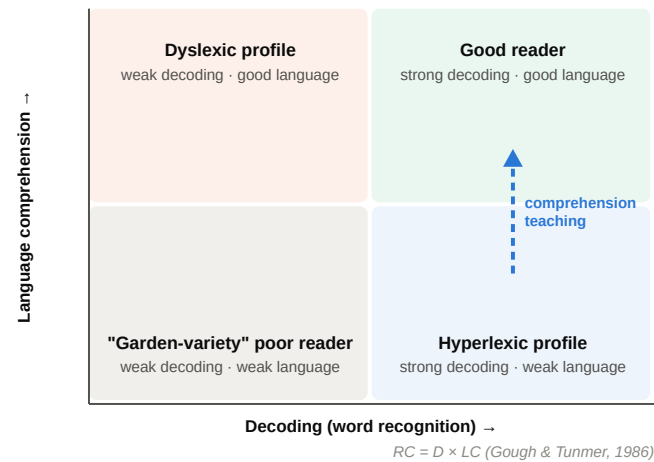
Definitions have drifted along three axes: *what* is advanced (single-word reading), *relative to what* (comprehension, vocabulary, or IQ), and *in whom* (any child, or only children with developmental conditions) [16] (Table 1). The most cited modern definition requires the co-occurrence of (1) reading skill advanced relative to comprehension or general intelligence, (2) early acquisition without explicit teaching, and (3) a strong orientation toward written material, generally in the context of a neurodevelopmental condition [18]. Clinical descriptions add typical signs: preference for books and letters over toys, spelling words aloud or tracing them in the air, teaching oneself to read, and difficulty answering *who* / *what* / *where* / *why* / *how* questions [27, 28].

**Table 1.** How the definition of hyperlexia has evolved.

| Source                                               | Definition / emphasis                                                                                                        |
|------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| Silberberg & Silberberg 1967 [1]                     | Specific word-recognition skill far above other abilities in young children.                                                 |
| Elliott & Needleman 1976 [5]                         | A distinct syndrome.                                                                                                         |
| Richman & Kitchell 1981; Cohen 1987, 1997 [6, 7, 29] | Variant or subgroup of developmental language disorder, not dyslexia.                                                        |
| Snowling & Frith 1986 [12]                           | “True” hyperlexia = failure to build large units of meaning in low-verbal fluent readers.                                    |
| Nation 1999 [15]                                     | Tail of normal variation in reading components plus compulsive print exposure.                                               |
| Grigorenko et al. 2003 [16]                          | Superability: unexpected single-word reading in a developmental disorder.                                                    |
| Ostrolenk et al. 2017 [18]                           | Reading > comprehension or IQ; early and untaught; strong orientation to print; usually with a neurodevelopmental condition. |
| Zhang & Joshi 2019 [19]                              | Good decoding with poor listening and reading comprehension (Simple View).                                                   |
| Practice literature [30]                             | A learning style, frequently with gestalt language processing.                                                               |

### 3.2 The Simple View of Reading

The Simple View holds that reading comprehension is the *product* of decoding and language comprehension [25]. It yields good readers and three kinds of poor readers — dyslexic (weak decoding, intact language), hyperlexic (strong decoding, weak language comprehension) and "garden-variety" (both weak) (Fig. 3) [31]. Because the terms multiply, even perfect decoding cannot compensate for weak language comprehension; hence the argument that hyperlexia is a genuine reading disorder requiring language support [32]. The meta-analysis by Zhang and Joshi supports the good-decoding / poor-listening-and-reading-comprehension profile, but found that only 66.22% of identified hyperlexics actually showed *precocious* decoding and that about 96% of studies sampled atypical populations, so definitions strongly shape results [19].



**Figure 3.** The Simple View of Reading as a map of reader profiles (conceptual, not data). Hyperlexia occupies the strong-decoding / weak-language quadrant; intervention aims to move language comprehension upward while preserving decoding.

### 3.3 Disability, superability, or learning style?

Grigorenko, Klin and Volkmar concluded that hyperlexia is best seen as a *superability* of a specific group of children with developmental disorders [16]. Nation argued that it sits at the tail of normal variation in phonological, orthographic and semantic processing, amplified by compulsive print exposure [15]. Kennedy's two cases supported an "asset rather than deficit" reading [33], whereas neuropsychological comparisons placed its essential feature in language impairment rather than dyslexia [7, 29]. Clinicians working with gestalt language processors describe it as a learning *style*: many hyperlexic children acquire language in chunks and learn best when language is modelled in writing [30]. These views are compatible: the reading engine is a strength; the language-to-meaning link is the target.

## 4 SUBTYPES AND DIFFERENTIAL DIAGNOSIS

Treffert distinguished three types (Table 2) [28, 34]. Type 1 describes typically developing precocious readers, exemplified by a preschooler with a reading age of 11.2 years at 4;2 and comprehension well above age level [35]. Type 2 is hyperlexia as a

splinter skill within autism, often with prodigious memory for dates, plates or trivia. Type 3 describes early readers with "autistic-like" traits that fade; misreading them as autistic can lead to inappropriate placement [34, 36, 37]. Treffert's broader work on savant syndrome estimates that up to one in ten autistic people have savant abilities, that about half of savants are autistic, that IQ can be superior, and that savant skills tend to grow from duplication through improvisation to creation [36].

**Table 2.** Treffert's three types of hyperlexia [28, 34].

| Type | Who                                       | Key features                                                                                                                       | Course and support                                                 |
|------|-------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|
| 1    | Typically developing early readers        | Reads very early, comprehension usually good                                                                                       | Gap closes as peers learn to read; no treatment needed [35]        |
| 2    | Autistic children                         | Letter/number fascination in infancy as a splinter skill; strong memory (dates, plates, scripts); social and language difficulties | Persistent; specialised education, speech and occupational therapy |
| 3    | Early readers with "autistic-like" traits | Interactive, affectionate, better eye contact; language and peer difficulties                                                      | Traits fade; mainstream classroom; good outcomes [37]              |

Within high-functioning hyperlexia, Richman and Wood identified two subtypes among 30 children: a language-learning-disorder pattern with good visual memory and phonetic errors, and a nonverbal-learning-disorder pattern with visuospatial weakness and sight-word errors [38]. Turner syndrome shows yet another form — broad "hyperdevelopment" of reading with *good* comprehension [39, 40]. Snowling and Frith found that more verbally able advanced decoders comprehended normally and proposed that "true" hyperlexia is marked by failure to build large units of meaning; they found no autistic/non-autistic differences [12]. Other cases show comprehension that is not unexpectedly deficient [10] or above average [41].

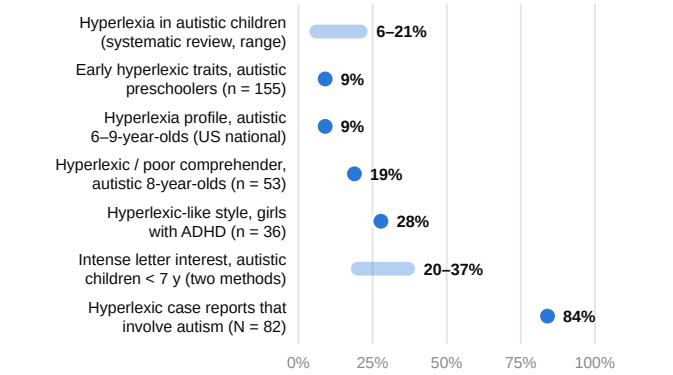
**Box 1 · Acquired hyperlexia in adults.** Neurologists also use "hyperlexia" for a different phenomenon: adults who, after lesions of the left medial frontal lobe, anterior cingulate or corpus callosum, compulsively read aloud irrelevant environmental text — trash bags, signs, newspapers — often with echolalia [42–45]. It is interpreted as bottom-up stimulus processing escaping top-down inhibition [46, 47]. It shares a name, not a developmental course, with childhood hyperlexia.

## 5 PREVALENCE AND CLINICAL ASSOCIATIONS

### 5.1 In autism

Estimates depend on criteria (Fig. 4). The systematic review of 82 cases and 912 group-study participants (315 hyperlexic) reported hyperlexia in 6–21% of autistic children, and 84% of published hyperlexic cases were autistic [18]. Using a 90th-percentile criterion for early letter/number naming, 9% of 155 autistic preschoolers showed early hyperlexic traits [48]. A nationally representative US sample of autistic 6–9-year-olds contained a hyperlexia achievement profile in 9% [49], and a Swedish population cohort found 10 of 53 autistic 8-year-olds (19%) with a

"hyperlexic/poor comprehender" profile [50]. Broader *interest* in letters is much commoner (20–37%; §6). Hyperlexia is significantly more frequent in pervasive developmental disorders than in other developmental diagnoses, with no sex difference [51]. Nation and colleagues' sample of 41 autistic children ranged from floor to ceiling on every reading measure, underlining heterogeneity [52].



**Figure 4.** Reported rates across studies. Rows measure different constructs and must not be compared as a trend: rows 1–5 are hyperlexic reading profiles, row 6 is *interest* in letters, and row 7 is the share of published hyperlexia cases that are autistic. Sources: Ostrolenk 2017; Solazzo 2021; Wei 2015; Åsberg Johnels 2019a,b; Ostrolenk 2024.

### 5.2 Beyond autism

A hyperlexic-like style was found in 10 of 36 girls with ADHD, who also had more autistic social-communication features [53]. More than two-thirds of young people with 22q11.2 deletion syndrome read above IQ expectations [54]. Case reports extend the list to Prader-Willi syndrome [55], West syndrome with reading of Japanese and Chinese characters at age three [56], tuberous sclerosis with hypermnnesia and hypercalculia in a non-speaking girl [57], specific language impairment [29], semantic-pragmatic disorder [58], and a gifted boy after early institutionalisation [59] (Table 3). Even typical high-school students fit a hyperlexic profile when reading a newly learned foreign language — "reading without meaning" is a normal early stage of L2 literacy [31].

**Table 3.** Conditions in which hyperlexia has been reported.

| Condition               | Hyperlexia evidence                                              |
|-------------------------|------------------------------------------------------------------|
| Autism                  | Most common context; 6–21% of autistic children [18]             |
| ADHD (girls)            | 10 of 36 with hyperlexic-like style; more autistic features [53] |
| 22q11.2 deletion        | > 2/3 read above IQ expectations [54]                            |
| Turner syndrome         | Reading above age and IQ with good comprehension [39]            |
| Prader-Willi            | Case report [55]                                                 |
| West syndrome           | Kanji, numbers, Roman letters read at 3 y [56]                   |
| Tuberous sclerosis      | Non-speaking girl with hyperlexia and hypercalculia [57]         |
| Tourette + PDD          | Chance co-occurrence $p = 3.39 \times 10^{-12}$ [60]             |
| Language disorder / SLI | Hyperlexia as SLI subgroup [6, 29]                               |
| Acquired (adults)       | Frontal/cingulate lesions, encephalopathy (Box 1)                |

### 5.3 Families, twins and genes

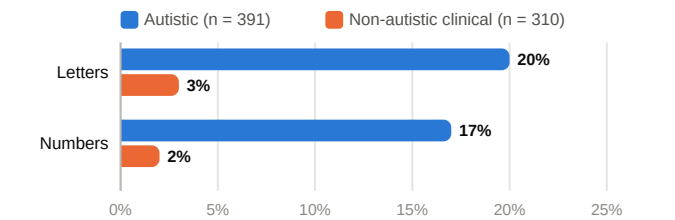
Family histories of 12 hyperlexic children revealed language, reading, writing and spelling disorders in male relatives and unusually high non-left-handedness [61]. Five North Dakotans had PDD, Tourette disorder and hyperlexia together; the authors computed the chance probability of this triple co-occurrence as  $3.39 \times 10^{-12}$  (about 1 in 295 billion) and inferred genetic linkage [60]. Identical twins concordant for autism and hyperlexia have been described twice [20, 62]. In 129 autistic children, hyperlexia clustered with a family history of affective disorder and of unusual intellectual achievement, suggesting a distinct subgroup [63]. Hyperlexic autistic children in one small comparison showed more persistent echolalia, better visual-motor skills and, in two cases, macrocephaly [64]. At the animal level, mice lacking the autism-risk gene SHANK3 show enhanced pitch discrimination, a reminder that heightened low-level perception may be heritable [65].

## 6 EARLY DEVELOPMENT: INTEREST IN LETTERS AND NUMBERS

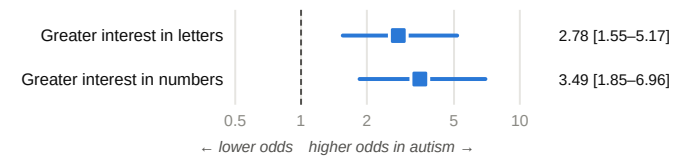
The largest study of early print interest reviewed all 701 children under 7 referred to a Montréal autism clinic over four years (391 autistic, 310 non-autistic) and interviewed caregivers of 313 children [21]. Clinician reports described an intense or exclusive interest in letters in 20% of autistic children versus 3% of clinical peers, and in numbers in 17% versus 2% (Fig. 5). Adjusted odds of a greater interest were 2.78 for letters and 3.49 for numbers (Fig. 6). When caregivers were asked directly, the proportion with intense letter interest rose to 37%.

Timing is the striking result: the median age of emergence was 30 months for autistic children, 28.5 months for typically developing children and 36 months for the non-autistic clinical group, and it did not depend on the child's level of oral language (Fig. 7). That interest arose while 24% of autistic children had no oral language and a further 37% used isolated words only (Fig. 8). Of 33 fully non-verbal autistic children, 8 could sing the alphabet

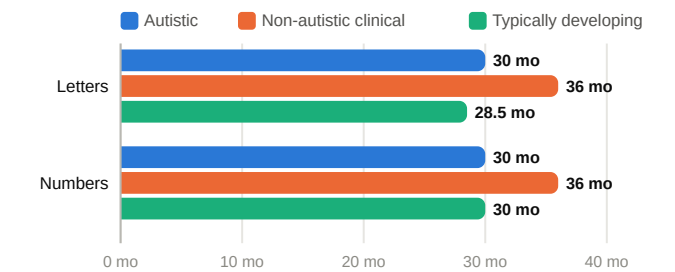
song, 9 named letters, 9 counted, 3 pretend-read, 2 read some words and 1 spelled some words. Twenty-five percent of typically developing children also had an intense interest in letters, but theirs usually appeared after speech and centred on shared books, whereas autistic children engaged alone and often refused adult intervention — a pattern that may lead parents to underestimate their competence [21]. A separate study reported that 29.7% of autistic 9-year-olds had a special interest in reading, suggesting persistence into childhood (cited in [21]).



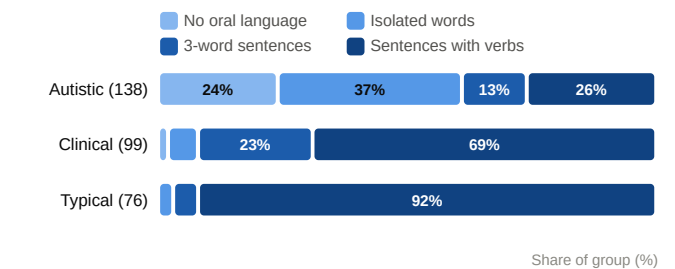
**Figure 5.** Children under 7 whose clinical report described an *intense or exclusive* interest in letters or numbers (Ostrolenk et al., 2024, Study 1).



**Figure 6.** Adjusted odds ratios (95% CI, log scale) for a greater level of interest in autistic vs non-autistic clinical children, controlling for age and assessing clinician (Ostrolenk et al., 2024).



**Figure 7.** Median age (months) at which interest in letters and in numbers emerged, caregiver interview (n = 138 autistic, 99 clinical, 76 typical). Autistic and typical children did not differ; the clinical group was later (letters HR = 0.73, p = .029; numbers HR = 0.63, p = .002).



**Figure 8.** Most complex oral language reported by caregivers in the same cohort. Letter interest emerged on schedule even though 61% of autistic children had no speech or single words only.

Early letter/number skill is not a bad sign. In 155 preschoolers it was associated with more repetitive behaviours but also with more social-oriented behaviour and better expressive and written communication one year later [48]. Six Brazilian boys whose

hyperlexia was noticed before 36 months all recognised letters and numbers, though most did not understand what they read [66]. The Montréal twins' language consisted only of letters and numbers until their fourth year, after which vocabulary broadened to full sentences and their perceptual interests "merged to serve the development of other skills" [20]. The authors frame hyperlexia as potentially the first step of a *non-social* pathway into language [20, 21].

## 7 COGNITIVE AND NEURAL MECHANISMS

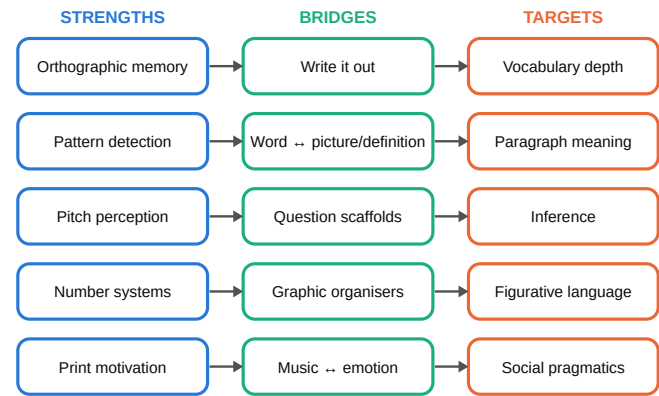
### 7.1 Routes from print to sound

Almost every mechanism study agrees that hyperlexic readers can reach pronunciation without meaning. Two cases read irregular words they could not define as accurately as words they could, evidence of a direct lexical route from orthography to phonology [67]; whole-word orthographic and phonological representations were acquired despite severe semantic impairment [68]; and girls with severe intellectual disability read words, non-words and texts accurately [69], as did a girl with an IQ of 58 [8]. Beyond that consensus, routes differ by child. Eight children in one study reached the lexicon by both visual-orthographic (preferred) and phonological routes and understood single words and sentences but not paragraphs; only three showed metalinguistic awareness [9]. Five others preferred the phonological route, resembling surface dyslexia [10]. Hyperlexic children retain sound–symbol pairings unusually well [70] and are faster and finer-grained at reading degraded "incomplete words" [11]. Preschoolers with autism and hyperlexia had advanced word reading and letter naming without matching phonological awareness or letter–sound knowledge — a non-phonological path [23] — and Chinese hyperlexic reading appears to map whole characters directly to sound [71]. In adolescents, outstanding readers had stronger lexical orthographic and phonological representations but no advantage in sublexical processing, rapid naming or memory [72]. The general architecture of single-word reading is nevertheless the same as in typical readers [73].

### 7.2 Perception first: EPF and veridical mapping

The Enhanced Perceptual Functioning model holds that perception has a larger, more autonomous role in autistic cognition [26]. A meta-analysis of imaging studies found more temporal, occipital and parietal and less frontal activity in autism across faces, objects and words [74]. *Veridical mapping* proposes that perceptual systems detect isomorphisms between structures — letters and sounds, pitches and note names, numbers and colours — explaining why hyperlexia, absolute pitch and synaesthesia co-occur and develop alike [22, 75]. Autistic children therefore follow a "chronologically inverted" path, mastering the code before the meaning, and make extended use of the visual word-form system [18]. Goldberg's older savant account emphasised intact declarative but impaired procedural memory [76]; clinical reviews stress visual memory, visual discrimination and motivation toward

visual material [77, 78]. A minimally verbal trilingual hyperlexic child processed recursive visual patterns like typical peers despite no recursion in speech [79].



Veridical mapping (Mottron 2013) links the strengths; teaching supplies the bridges.

**Figure 9.** A strength-to-target model synthesised from the reviewed literature (conceptual). Perception-first strengths are routed through print-based bridges toward the comprehension skills that lag.

7.3 Neuroimaging

The only fMRI study of developmental hyperlexia examined a 9-year-old reading six years ahead of his age. During covert reading he over-activated left inferior frontal and superior temporal cortex (vs both age- and reading-age-matched controls) and right inferior temporal sulcus (vs reading-age controls). Hyperlexic reading thus draws simultaneously on left-hemisphere phonological and right-hemisphere visual systems — the mirror image of dyslexic hypoactivation of left superior temporal cortex [17]. Orthographic-route accounts from Spanish-speaking cases converge on the same idea [80].

7.4 Memory and profiles

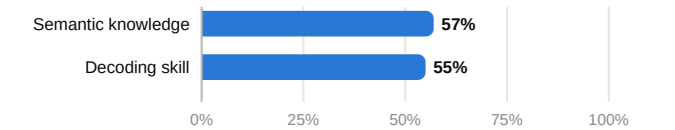
Hyperlexic autistic boys followed for 7–17 years had unusually good visual and auditory memory and a large stored written vocabulary despite limited speech [78]. SLI + hyperlexia was distinguished by stronger visual/spatial memory, while verbal memory fell as semantic load rose [29]. Children with West syndrome showed the opposite, excellent auditory memory with weak visual retention [56]. Hypergraphia may accompany hyperlexia [81], and IQ varies enormously — from severe disability to very superior — with prognostic implications [51, 78, 82].

8 READING COMPREHENSION

8.1 What predicts understanding

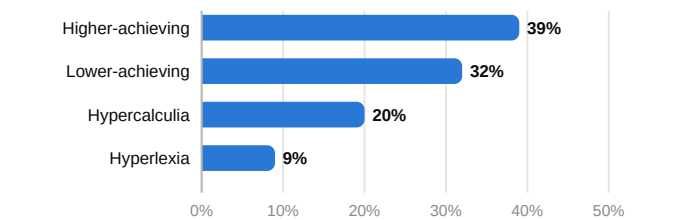
Across 36 studies, autistic readers scored 0.7 SD below controls in comprehension; semantic knowledge explained 57% and decoding 55% of the variance, and highly social texts were harder than non-social ones — having autism alone did not predict difficulty [83] (Fig. 10). A later meta-analysis of 26 studies (ages 6–18) found similar mean correlations of comprehension with word reading ( $r = .65$ ) and with oral language ( $r = .61$ ) [84]. In poor

comprehenders generally, a subset with lower general ability shows a hyperlexic profile in which accuracy, not comprehension, is the surprise [85].

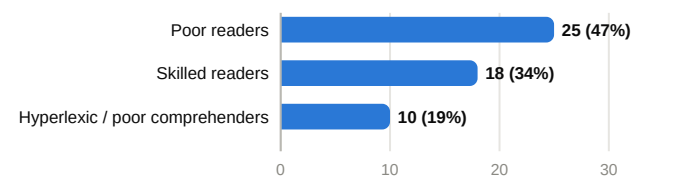


**Figure 10.** Variance in reading comprehension explained by single predictors across 36 studies of autistic readers (Brown et al., 2013). Overall comprehension difference:  $g = -0.7$  SD; socially loaded texts were harder than non-social ones.

Development matters. In the nationally representative Wei et al. sample, every achievement profile lost ground in passage comprehension over time and the hyperlexia group improved most slowly in conversation (Fig. 11) [49]. In the Swedish cohort, oral-language weaknesses already visible at age 3 predicted who became "hyperlexic/poor comprehenders" at 8 (Fig. 12) [50]. Early assessment can be done online: a six-subtest multimedia tool (decoding, homographs, auditory/visual vocabulary and sentence comprehension) separated autistic from typical children with up to 92% accuracy [86].



**Figure 11.** Reading/math achievement profiles in a nationally representative US sample of autistic children aged 6–9 (Wei et al., 2015). All four profiles lost ground in passage comprehension over time.



**Figure 12.** Reading profiles at age 8 in a Swedish population cohort of autistic children identified by screening before age 3 ( $n = 53$ ; Åsberg Johnels et al., 2019b). Oral-language weakness at 3 predicted the hyperlexic profile.

8.2 From words to paragraphs

Comprehension breaks down as units grow. Children who understood single words and sentences failed with paragraphs [9]. Low-verbal fluent readers read word-by-word, failed to use sentence context to pronounce homographs and did not draw on general knowledge to answer questions about stories [12]. The core difficulty, in Shields' words, is integrating semantic information with knowledge of the world [58].

### 8.3 Inference, figurative language and social text

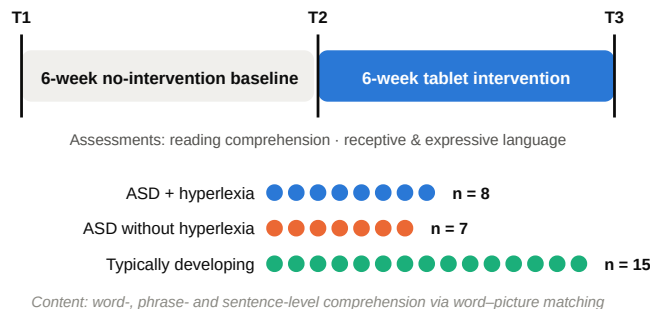
Autistic children struggle most with inferences that require context, especially when mind-reading is involved, and with explaining how they used context [87]. Idioms are often read literally; comprehension depends on inferencing and is helped by transparency, familiarity and context [88]. An ALE meta-analysis of six fMRI studies (95 autistic, 98 non-autistic participants) found a shared figurative-language network but consistent hypoactivation of left superior and middle temporal gyri in autism [89]. In 81 autistic 8–16-year-olds followed over three waves, growth in narrative retelling and inference related to reading-comprehension outcomes [90].

### 8.4 Adolescence

Adolescent evidence is thin but consistent. In 45 autistic adolescents (mean 14.9 years), task-switching predicted comprehension indirectly through decoding [91]; middle-school students improved with vocabulary and main-idea instruction and valued choosing their own texts [92]; and a hyperlexic student followed into high-school Spanish was strongest on pronunciation, word recognition and spelling [93]. Twice-exceptional autistic students thrive with strength- and interest-based teaching [94], and their parents describe high school as a time when intellectual strengths and social demands diverge sharply [95].

## 9 INTERVENTION

Table 4 summarises the intervention evidence. The oldest demonstration remains one of the most useful: written prompts rapidly increased a hyperlexic child's appropriate spoken responses in natural settings, with maintenance and generalisation [70]. The Scaffolding Interrogative Method guides children through structured questions; comprehension rose in each intervention phase and comprehension age increased while reading age did not, closing the hyperlexic gap (secondary summaries report gains of 19.4–36.3% in three months) [96]. The McGill tablet program trained comprehension at word, phrase and sentence level through word–picture matching with parent support; after a six-week baseline and six weeks of intervention, children with autism and hyperlexia gained significantly more in reading comprehension than typical peers ( $p = .023$ ), and all groups gained in receptive language (Fig. 13) [24, 97].



**Figure 13.** Design of the McGill parent-supported tablet intervention (Macdonald, Luk & Quintin, 2022; N = 30). Reading-comprehension gains were significantly larger for ASD + hyperlexia than for typical peers ( $p = .023$ ).

**Table 4.** Intervention evidence relevant to hyperlexic learners.

| Study                  | Design · N                     | Approach                                            | Outcome                                                                 |
|------------------------|--------------------------------|-----------------------------------------------------|-------------------------------------------------------------------------|
| Kistner 1988 [70]      | Single case                    | Written prompts for speech                          | Rapid gains in appropriate verbal responses; maintained and generalised |
| Ng & Chia 2014 [96]    | Repeated baseline–intervention | Scaffolding Interrogative Method                    | Comprehension age ↑ without reading-age change                          |
| Macdonald 2022 [24]    | Baseline vs intervention · 30  | Tablet word–picture matching, parent-supported      | Comprehension ↑ for ASD+HPL vs TD ( $p = .023$ ); receptive language ↑  |
| El Zein 2014 [98]      | Synthesis · 12 studies         | Question generation, organisers, prediction, cueing | Improved comprehension                                                  |
| Lee 2025 [99]          | Meta-analysis · 5 SCED         | Pictorial/graphic representation                    | Tau-U = 0.85                                                            |
| Solis 2021 [92]        | Single-case · 5 adolescents    | Vocabulary + main idea with text choice             | Upward, stable trends; valued choice                                    |
| Stein 2015 [59]        | Case · 1                       | Floortime + pragmatic group SLT                     | Friendships and sociability improved                                    |
| Geretsegger 2022 [100] | Cochrane · 26 studies          | Music therapy                                       | Global improvement RR 1.22                                              |

For school-age autistic readers, effective components include question generation, graphic organisers, prediction, anaphoric cueing, explicit instruction and grouping [98], and pictorial or graphic representations yield moderate-to-strong effects (Tau-U = 0.85) [99]. A gifted hyperlexic boy improved markedly with Floortime plus social/pragmatic group speech therapy [59].

Clinical consensus recommends speech-language therapy, occupational therapy, play-based behavioural approaches and psychological support as needed, together with thoughtful school placement by type [27, 28]. Practitioners converge on print-first practice: model language in writing; keep letters and keyboards available (including keyboard-based AAC); turn on subtitles [30]; use visual supports for schoolwork, comprehension, communication, self-regulation, executive function, social skills and self-advocacy — "include the written word as much as possible" [101]; exploit rote learning and special interests; break tasks down; and "when in doubt, write it out" [102]. For *wh*-questions, present question and answer together, have the child read both, then fade the answer.

## 10 ALLIED STRENGTHS: NUMBERS, MUSIC AND LANGUAGES

### 10.1 Hypernumeracy

Numbers travel with letters. Autistic children had 3.49 times the odds of a stronger interest in numbers [21]; a hypercalculia achievement profile characterised 20% of a national autistic sample [49]; and case reports pair hyperlexia with hypercalculia and hypermnnesia [57, 103]. Reading of number words and Arabic numerals can even dissociate from reading of other words [40].

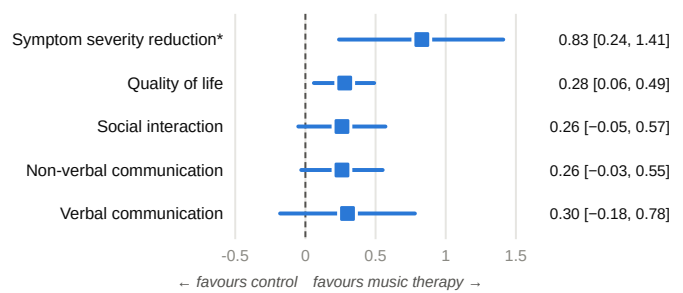
Large-number naming offers a bilingual twist (Table 5): English uses the *short scale* (a new name every  $10^3$ ), while French — including Québec usage — uses the *long scale* (a new name every  $10^6$ , with *-iard* names between). An English nonillion is  $10^{30}$ ; a French *nonillion* is  $10^{54}$ .

**Table 5.** Large-number names in the two languages of the intended player.

| Name                      | English (short scale)       | French / Québec (long scale)                     |
|---------------------------|-----------------------------|--------------------------------------------------|
| million                   | $10^6$                      | $10^6$                                           |
| milliard                  | —                           | $10^9$                                           |
| billion                   | $10^9$                      | $10^{12}$                                        |
| billiard                  | —                           | $10^{15}$                                        |
| trillion                  | $10^{12}$                   | $10^{18}$                                        |
| quadrillion               | $10^{15}$                   | $10^{24}$                                        |
| quintillion               | $10^{18}$                   | $10^{30}$                                        |
| sextillion                | $10^{21}$                   | $10^{36}$                                        |
| septillion                | $10^{24}$                   | $10^{42}$                                        |
| octillion                 | $10^{27}$                   | $10^{48}$                                        |
| <b>nonillion</b>          | <b><math>10^{30}</math></b> | <b><math>10^{54}</math></b>                      |
| decillion                 | $10^{33}$                   | $10^{60}$                                        |
| rule for <i>n</i> -illion | $10^{3n+3}$                 | $10^{6n}$ (and <i>n</i> -illiard = $10^{6n+3}$ ) |

### 10.2 Music

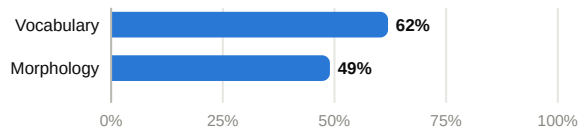
Music is a relative strength in autism: pitch perception, musical memory and identification of music-evoked emotion [104]. Absolute pitch occurs in an estimated 5–11% of autistic people [105], has been documented in a 4-year-old [106], and in one autistic adult extended to naming the pitch of spoken words far better than controls who also had absolute pitch [107]. Autistic children recognised basic emotions in music with a relative advantage, while matching peers on faces and voices [108] — making music a bridge to emotional vocabulary. Physiological responses to music-evoked emotion are reduced in some autistic youth, with age-related differences between 8–11 and 16–18 years [109]. The 2022 Cochrane review (26 studies, 1165 participants) found moderate-certainty benefits of music therapy for global improvement, quality of life and symptom severity, and no clear immediate effect on social interaction or communication (Fig. 14) [100]. Veridical mapping predicts that staff notation — a second orthography mapping symbols to pitches — should be learned with the same ease as print [22, 75].



**Figure 14.** Music therapy vs placebo therapy or standard care, standardised mean differences (95% CI) immediately post-intervention (Geretsegger et al., 2022; Cochrane). \*Reported as SMD  $-0.83$ ; sign reversed so that right = benefit. Global improvement: RR 1.22 [1.06, 1.40]; certainty moderate for severity, quality of life and global improvement, low/very low for the rest.

### 10.3 Bilingualism and multilingual hyperlexia

Hyperlexia crosses scripts: it has been reported in Chinese, Japanese, Kannada, Italian, Spanish, Catalan, Portuguese and English [56, 66, 69, 71, 79, 80, 110]. In a Kannada–English bilingual the decoding–comprehension gap appeared in *both* languages [110]. For autistic children, bilingual exposure is not detrimental: in a Montréal study the current amount of exposure to a language explained 62% of variance in vocabulary and 49% in morphology in autistic and typical children alike (Fig. 15) [111], and bilingual autistic children scored in the average monolingual range for receptive vocabulary [112]. Morphology, including shared Latin and Greek roots, is worth targeting explicitly, since it was the weakest meta-linguistic skill in a hyperlexic Chinese reader [71].



**Figure 15.** Variance explained by the current amount of exposure to a language in school-age children with and without autism in Montréal ( $n = 30$  autistic, 47 typical; Gonzalez-Barrero & Nadig, 2018). Exposure mattered as much for autistic children as for typical ones.

## 11 OUTCOMES ACROSS ADOLESCENCE AND ADULTHOOD

Long-term data are sparse but mostly encouraging. Hyperlexia was one of four variables — with known aetiology, Tourette disorder and age — predicting higher IQ and receptive/expressive language in North Dakota's 59 children with PDD, and was proposed as a marker of improvement [14, 113]. Developmental outcomes of hyperlexic children in a referred sample were comparable to those of peers [51]. Type 3 features fade, sometimes dramatically [34, 37]. Preschool hyperlexic traits predicted better expressive and written communication a year later [48], and hyperlexic twins progressed from letter-and-number-only language to full sentences [20]. The caution is comprehension: passage comprehension falls further behind with age unless it is taught [49]. Special skills can be channelled toward employment [103] and, according to Treffert, tend to deepen toward creativity rather than disappear [36].

# 12 DESIGN IMPLICATIONS: A BILINGUAL, MUSIC-INTEGRATED GAME SUITE

The evidence yields nine design principles, each applied throughout the accompanying 25-game suite (Table 6):

**P1 Lead with print.** Instructions, feedback and social scripts are written first; written prompts increase functional language [30, 70, 101].

**P2 Bridge code to meaning.** Every word found or typed is paired with a definition or picture; semantic knowledge is the largest predictor of comprehension [24, 83].

**P3 Climb the unit ladder.** Activities progress from words to sentences to paragraphs, where comprehension breaks down [9, 12].

**P4 Ask, then prove.** Questions require clicking the sentence that proves an answer (question generation, scaffolding) [96, 98].

**P5 Make meaning visible.** Graphic organisers, timelines, emotion meters and diagrams [99].

**P6 Teach the implicit explicitly.** Inference, idioms (English and Québécois), tone, sarcasm and perspective [83, 87, 88].

**P7 Harness allied systems.** Big numbers, codes and alphabets, staff notation and ear training ride on veridical mapping and music strengths [21, 22, 104].

**P8 Two languages, equal status.** Every game runs in English and Québec French; exposure drives vocabulary [111].

**P9 Challenge and choice.** Difficulty pitched at a strong 17-year-old, free choice among games, optional relaxed (untimed) mode for regulation [92, 94, 109].

**Table 6.** The 25-game suite and the evidence each game draws on.

| Game (EN / FR)                                        | What the player does                                                      | Skill targeted                                                    | Principle · evidence                 |
|-------------------------------------------------------|---------------------------------------------------------------------------|-------------------------------------------------------------------|--------------------------------------|
| 1 <b>Lexicon Lock</b><br><i>Cadenas lexical</i>       | Deduce a hidden 5-letter word in 6 tries                                  | Orthographic reasoning; meaning card after each word              | P2, P8 · orthographic asset [11, 33] |
| 2 <b>Wheel of Words</b><br><i>Roue des mots</i>       | Spin, guess letters, solve proverbs & expressions                         | Phrase-level meaning from partial print; idiom explained on solve | P3, P6 [11, 88]                      |
| 3 <b>Cipher Station</b><br><i>Station Chiffre</i>     | Crack Caesar, Atbash, Vigenère & substitution codes with frequency charts | Systemising, letter-pattern mapping                               | P7 · veridical mapping [21, 22]      |
| 4 <b>Word Invaders</b><br><i>Envahisseurs de mots</i> | Type words to blast ships; bosses need the word that fits a definition    | Automaticity → semantic retrieval                                 | P2 [24, 83]                          |
| 5 <b>Spelling Hive</b><br><i>Ruche à mots</i>         | Build words from 7 letters around a centre letter                         | Orthography + vocabulary depth (definitions)                      | P2 [83]                              |
| 6 <b>Grid Blitz</b><br><i>Blitz de grille</i>         | Trace words in a letter grid against the clock                            | Rapid orthographic search in both lexicons                        | P8 [111]                             |
| 7 <b>Word Ladder</b><br><i>Échelle de mots</i>        | Change one letter at a time from start to goal                            | Orthographic manipulation, planning                               | P2, P9                               |

| Game (EN / FR)                                                   | What the player does                                                            | Skill targeted                         | Principle · evidence |
|------------------------------------------------------------------|---------------------------------------------------------------------------------|----------------------------------------|----------------------|
| 8 <b>Letter Cascade</b><br><i>Cascade de lettres</i>             | Clear falling letters by spelling words                                         | Speed, flexibility, task-switching     | P9 [91]              |
| 9 <b>Crossword Forge</b><br><i>Forge à mots croisés</i>          | Solve generated crosswords from clues                                           | Definition → word retrieval            | P2 [83, 84]          |
| 10 <b>Four Groups</b><br><i>Quatre groupes</i>                   | Sort 16 words into 4 hidden categories                                          | Semantic categories, multiple meanings | P2, P6 [12]          |
| 11 <b>Idiom Detective</b><br><i>Détective des expressions</i>    | Crack literal vs figurative meanings (English idioms & expressions québécoises) | Figurative language, context           | P6 [88, 89]          |
| 12 <b>Mystery Files</b><br><i>Dossiers mystère</i>               | Read case files, answer wh-questions, click the evidence                        | Paragraph inference with proof         | P3, P4 [9, 87, 96]   |
| 13 <b>Social Signals</b><br><i>Signaux sociaux</i>               | Decode feelings, tone and sarcasm in chats and scenes                           | Emotion vocabulary, pragmatics         | P5, P6 [83, 108]     |
| 14 <b>Conversation Quest</b><br><i>Quête de conversation</i>     | Choose replies in branching teen scenarios                                      | Written social scripts                 | P1, P6 [59, 70]      |
| 15 <b>Story Sequencer</b><br><i>Chrono-récit</i>                 | Rebuild scrambled stories and cause-effect chains                               | Narrative structure                    | P3, P5 [90, 99]      |
| 16 <b>Proofreader Pro</b><br><i>Correcteur pro</i>               | Find errors where meaning decides spelling                                      | Homophones, agreement, context         | P2 [12]              |
| 17 <b>Root Lab</b><br><i>Labo des racines</i>                    | Build words from Greek/Latin roots shared by EN and FR                          | Morphology, cross-language transfer    | P2, P8 [71, 111]     |
| 18 <b>Nonillion Navigator</b><br><i>Navigateur de nonillions</i> | Name, convert and compare huge numbers; short vs long scale                     | Hypernumeracy as a bridge to language  | P7, P8 [21, 49]      |
| 19 <b>Sequence Lab</b><br><i>Labo des suites</i>                 | Find the rule in number/letter sequences                                        | Pattern detection                      | P7 [26, 74]          |
| 20 <b>Signal Decoder</b><br><i>Décodeur de signaux</i>           | Learn Morse, Braille, NATO, Greek and binary                                    | New orthographies                      | P7 [56, 79]          |
| 21 <b>Staff Speller</b><br><i>Portée magique</i>                 | Read notes on the staff that spell words; letters ↔ solfège                     | Music notation as orthography          | P7, P8 [75, 105]     |
| 22 <b>Pianissimo</b><br><i>Pianissimo</i>                        | Play falling-note melodies with scrolling lyrics                                | Music strength, reading-while-singing  | P7 [100, 104]        |
| 23 <b>Ear Trainer</b><br><i>Oreille d'or</i>                     | Identify notes, intervals and chord moods                                       | Pitch perception, music-emotion words  | P5, P7 [105, 108]    |
| 24 <b>Definition Snake</b><br><i>Serpent des définitions</i>     | Eat letters in order to spell the word a clue defines                           | Definition → spelling                  | P2 [24]              |
| 25 <b>Grammar Racer</b><br><i>Course grammaticale</i>            | Steer into the lane with the word that fits                                     | Sentence-level meaning                 | P3 [12, 24]          |

## 13 LIMITATIONS

Most hyperlexia studies are single cases or small groups; definitions and measures vary so much that pooling is inappropriate [19]. Adolescent and adult data are scarce. Some older and non-English records could only be summarised from bibliographic data, and a few practice sources are non-peer-reviewed. The SIM percentages come from secondary summaries. This review was compiled by an AI research assistant from publicly indexed abstracts and full texts; it has not been peer-reviewed and is not medical advice — decisions about assessment and therapy belong with qualified clinicians who know the young person.

## 14 CONCLUSIONS

Sixty years of research converge on a clear picture. Hyperlexia is a powerful, perception-driven print engine that often arrives before speech, frequently travels with numbers and music, and is most common — but not exclusive — in autism. Its limitation is not reading but *meaning-making*: vocabulary depth, inference, figurative and social language. The best-supported strategy is therefore to write everything out, use print to teach meaning, prove answers from the text, and let adjacent passions — nonillions, ciphers, sonatas, two languages — carry the learner toward the skills that are harder to reach.

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## APPENDIX A · ANNOTATED EVIDENCE TABLE (ALL 120 SOURCES)

Every source consulted, with its design, sample and unique contribution. † = abstract not indexed; summarised from the bibliographic record. Numbers in the first column match the reference list.

| Ref                                                           | Year | Source                        | Design · n                     | Unique contribution                                                                                                                                                                                                                                                  |
|---------------------------------------------------------------|------|-------------------------------|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>A1 · Records on hyperlexia (Europe PMC title/abstract)</b> |      |                               |                                |                                                                                                                                                                                                                                                                      |
| [1]                                                           | 1967 | Silberberg & Silberberg †     | Theory / commentary<br>—       | First scientific description; coined the term for precocious, specific word-recognition skill in young children.                                                                                                                                                     |
| [2]                                                           | 1968 | Niensted †                    | Theory / commentary<br>—       | Early commentary questioning whether hyperlexia is an educational problem ("disease") or a talent.                                                                                                                                                                   |
| [3]                                                           | 1972 | Mehegan & Dreifuss †          | Case / series series           | Framed hyperlexia as exceptional reading in children with brain damage/developmental delay.                                                                                                                                                                          |
| [4]                                                           | 1973 | Huttenlocher & Huttenlocher † | Case / series series           | Early neurological case series of children with hyperlexia.                                                                                                                                                                                                          |
| [5]                                                           | 1976 | Elliott & Needleman †         | Theory / commentary<br>—       | Proposed hyperlexia as a distinct syndrome.                                                                                                                                                                                                                          |
| [6]                                                           | 1981 | Richman & Kitchell †          | Group study<br>—               | Positioned hyperlexia within developmental language disorder rather than reading disorder.                                                                                                                                                                           |
| [11]                                                          | 1982 | Cobrinik                      | Group study<br>HPL vs controls | Hyperlexic children outperformed controls at deciphering words with deleted letter cues: faster, finer-grained, reliant on the whole visual array; linked to right-hemisphere visual processing.                                                                     |
| [114]                                                         | 1982 | Fontenelle & Alarcon          | Case / series 7                | Seven developmentally delayed children with precocious word recognition; discussed how the pattern may disrupt acquisition of communicative modalities.                                                                                                              |
| [115]                                                         | 1982 | Healy et al. †                | Group study<br>—               | Influential group study characterising hyperlexic children (bibliographic record only).                                                                                                                                                                              |
| [82]                                                          | 1983 | Graziani et al. †             | Group study<br>—               | Reported wide IQ variability and its prognostic meaning in hyperlexia.                                                                                                                                                                                               |
| [9]                                                           | 1984 | Goldberg & Rothermel          | Case / series 8                | Eight hyperlexic children read at grade 4–6 level; used both visual-orthographic (preferred) and phonological routes; understood single words and sentences but not paragraphs; only 3/8 showed metalinguistic awareness.                                            |
| [8]                                                           | 1984 | Siegel                        | Case / series 1                | Girl with WISC-R IQ 58 but advanced reading; read sentences she did not understand; argued reading can proceed via visual/phonological routes without semantic or syntactic processing.                                                                              |
| [78]                                                          | 1984 | Whitehouse & Harris           | Group study 20                 | Twenty hyperlexic autistic boys followed 7–17 years: compulsion to decode without comprehension; IQ from severe disability to very superior; unusually good visual and auditory memory; large stored written vocabulary.                                             |
| [81]                                                          | 1985 | Burd & Kerbeshian             | Case / series 1                | Co-occurrence of hyperlexia and hypergraphia (compulsive writing); hyperlexia proposed as a marker to guide educational programming.                                                                                                                                 |
| [13]                                                          | 1985 | Burd et al. †                 | Group study statewide          | Statewide (North Dakota) estimate of hyperlexia among children with PDD (bibliographic record only).                                                                                                                                                                 |
| [69]                                                          | 1986 | Cossu & Marshall              | Case / series 2                | Two Italian girls (12.5 and 18.6 y) with severe intellectual disability read words, non-words and texts accurately; reading/writing as transcoding can develop independently of other cognitive systems.                                                             |
| [61]                                                          | 1986 | Healy & Aram                  | Group study 12 families        | Family histories of 12 hyperlexic children: familial tendency to language, reading, writing and spelling disorders in male relatives; unusually high non-left-handedness.                                                                                            |
| [12]                                                          | 1986 | Snowling & Frith              | Group study 4 expts            | More verbally able advanced decoders comprehended normally; low-verbal readers read word-by-word, did not use general knowledge. Proposed that impaired large-unit meaning marks "true" hyperlexia; no autistic vs non-autistic differences, so not autism-specific. |
| [113]                                                         | 1987 | Burd et al. †                 | Group study<br>—               | Raised hyperlexia as a possible positive prognostic marker in PDD (bibliographic record only).                                                                                                                                                                       |
| [7]                                                           | 1987 | Cohen et al.                  | Case / series 5                | Five clinic patients: the primary deficit is comprehension of spoken and written language, i.e., a language disorder rather than dyslexia.                                                                                                                           |
| [76]                                                          | 1987 | Goldberg                      | Review<br>—                    | Review: hyperlexic readers use phonological and orthographic routes and derive meaning mainly from single words; savant theory of impaired procedural but intact declarative memory.                                                                                 |
| [35]                                                          | 1987 | Pennington et al.             | Case / series 1                | Typically developing boy with reading age 9.3 at 2;11 and 11.2 at 4;2, with comprehension above age level: precocious reading does not require pathology (cf. Treffert Type 1).                                                                                      |
| [10]                                                          | 1987 | Welsh et al.                  | Case / series 5                | Five boys: reading precocious relative to IQ, but comprehension not unexpectedly deficient; phonological route preferred; pattern resembled surface dyslexia.                                                                                                        |
| [60]                                                          | 1988 | Burd & Kerbeshian             | Case / series 5                | Five people in North Dakota with PDD + Tourette + hyperlexia; chance co-occurrence probability computed at $3.39 \times 10^{-12}$ , suggesting genetic linkage.                                                                                                      |
| [14]                                                          | 1988 | Fisher et al.                 | Group study 59                 | Across North Dakota's 59 children with PDD, hyperlexia was one of four predictors (with known aetiology, Tourette, age) of higher IQ and receptive/expressive language.                                                                                              |

| Ref   | Year | Source                       | Design · n                    | Unique contribution                                                                                                                                                                                                                        |
|-------|------|------------------------------|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [116] | 1988 | Ichiba †                     | Case / series<br>—            | Japanese neuropsychological/EEG study proposing a "benign hyperlexia" form (bibliographic record only).                                                                                                                                    |
| [70]  | 1988 | Kistner et al.               | Group study<br>2 studies      | Hyperlexic children retain sound/symbol associations unusually well (not just exposure). Written prompts rapidly increased appropriate verbal responses in natural settings, with maintenance and generalisation.                          |
| [62]  | 1988 | Smith & Bryson               | Case / series<br>2            | Identical twins concordant for autism and hyperlexia but differing in profile; highlights the autism–hyperlexia relationship.                                                                                                              |
| [55]  | 1989 | Burd & Kerbeshian †          | Case / series<br>—            | Reported hyperlexia in Prader-Willi syndrome (bibliographic record only).                                                                                                                                                                  |
| [117] | 1989 | Lachal                       | Theory / commentary<br>—      | French clinical analysis of hyperlexia as autism coupled with a compulsion to read or write and difficulty with self-expression.                                                                                                           |
| [56]  | 1990 | Ichiba                       | Case / series<br>2            | Two children with West syndrome read Japanese and Chinese characters, numbers, Roman letters and trademarks at 3 y; fluent intonation but impaired comprehension; excellent auditory memory.                                               |
| [58]  | 1991 | Shields                      | Theory / commentary<br>—      | Hyperlexia in semantic-pragmatic disorder reflects difficulty integrating semantic information with world knowledge; poor inferential meaning, metaphor and humour.                                                                        |
| [47]  | 1992 | Bindschaedler & Assal        | Acquired (adult)<br>review    | Adult neurology review placing acquired hyperlexia among environment-driven behaviours (utilisation, echolalia, hypergraphia).                                                                                                             |
| [118] | 1993 | Patti & Lupinetti †          | Case / series<br>1            | Hyperlexia described as a savant skill in an autistic individual (bibliographic record only).                                                                                                                                              |
| [64]  | 1993 | Tirosh & Canby               | Group study<br>5 + 5          | Hyperlexic autistic children showed more persistent echolalia, superior visual-motor performance; 2 had macrocephaly. Conclusion: part of the autism continuum, not a separate syndrome.                                                   |
| [39]  | 1996 | Temple & Carney              | Group study<br>TS vs controls | Girls with Turner syndrome read above age/IQ expectations with <b>good</b> comprehension and strength in both lexical and phonological routes: hyperlexia need not imply poor comprehension.                                               |
| [29]  | 1997 | Cohen et al.                 | Group study<br>46 + 16        | SLI+hyperlexia vs SLI: the essential feature is language impairment; better visual/spatial memory drives word recognition and spelling; verbal memory falls as semantic load rises.                                                        |
| [68]  | 1997 | Glosser et al.               | Case / series<br>1            | Orthographic and phonological whole-word representations can be acquired and retrieved without a functional link to semantic memory.                                                                                                       |
| [44]  | 1997 | Vuilleumier et al.           | Acquired (adult)<br>1         | Adult post-anoxic encephalopathy with compulsive environment-driven behaviours including hyperlexia.                                                                                                                                       |
| [15]  | 1999 | Nation                       | Review<br>—                   | Hyperlexia as part of normal variation in reading, explained by individual differences in phonological, orthographic and semantic processing, memory and print exposure; compulsive preoccupation may be crucial; connectionist framework. |
| [93]  | 2000 | Sparks & Artzer              | Case / series<br>3            | Hyperlexic student followed into high school Spanish: stronger on pronunciation, word recognition and spelling than on listening, speaking, writing; FL decoding may be modular.                                                           |
| [42]  | 2000 | Suzuki et al.                | Acquired (adult)<br>1         | 69-year-old with left medial frontal lesion compulsively read irrelevant environmental words during conversation (acquired hyperlexia with echolalia).                                                                                     |
| [46]  | 2000 | Yamadori                     | Acquired (adult)<br>theory    | Acquired hyperlexia explained as bottom-up stimulus processing overpowering top-down inhibition.                                                                                                                                           |
| [63]  | 2002 | DeLong et al.                | Group study<br>129            | In 129 autistic children, hyperlexia clustered with familial affective disorder, unusual family intellectual achievement and medication response, suggesting a genetically distinct subgroup.                                              |
| [51]  | 2002 | Grigorenko et al.            | Group study<br>80             | No sex difference in frequency; hyperlexia significantly more frequent in PDD than non-PDD diagnoses; IQ range and outcomes comparable to non-hyperlexic peers.                                                                            |
| [85]  | 2002 | Nation et al.                | Group study<br>25 + 24        | Most poor comprehenders had verbal-domain weaknesses; a subset with below-average general ability showed a hyperlexic profile where accuracy was surprisingly good.                                                                        |
| [38]  | 2002 | Richman & Wood               | Group study<br>30             | Two subtypes of high-functioning hyperlexia: (1) language-learning disorder with good visual memory and phonetic errors; (2) nonverbal-learning-disorder pattern with visuospatial deficits and sight-word errors.                         |
| [16]  | 2003 | Grigorenko et al.            | Review<br>—                   | Review of six controversies; concluded hyperlexia is a superability of a specific group with developmental disorders rather than a disability in the general population.                                                                   |
| [33]  | 2003 | Kennedy                      | Case / series<br>2            | Two distinct pathways to superior word recognition, both with specialised orthographic processing; supports an asset rather than deficit analysis.                                                                                         |
| [77]  | 2003 | Martos-Pérez & Ayuda-Pascual | Review<br>—                   | Links hyperlexia to autistic strengths in visual memory, visual discrimination and motivation toward visual material; "an island of ability".                                                                                              |
| [17]  | 2004 | Turkeltaub et al.            | Case / series<br>1 + controls | 9-year-old reading 6 years ahead: hyperactivation of left inferior frontal and superior temporal cortex and right inferior temporal sulcus; mirror image of dyslexic hypoactivation.                                                       |
| [45]  | 2006 | Díaz-Sepúlveda et al. †      | Acquired (adult)<br>1         | Acquired hyperlexia as a sign of Hashimoto's encephalopathy (bibliographic record only).                                                                                                                                                   |
| [52]  | 2006 | Nation et al.                | Group study<br>41             | In 41 autistic children, word/nonword reading were average but comprehension impaired, with floor-to-ceiling variability; some showed a hyperlexic profile.                                                                                |

| Ref   | Year | Source                      | Design · n                    | Unique contribution                                                                                                                                                                                                |
|-------|------|-----------------------------|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [80]  | 2006 | Talero-Gutierrez            | Case / series<br>2            | Two autistic Spanish-speaking children followed 8 years: self-taught reading before 5, minimal comprehension, obsessional reading; orthographic route proposed.                                                    |
| [40]  | 2006 | Temple                      | Review<br>—                   | Hyperlexia takes several forms, including broad hyperdevelopment (Turner); reading of number words and Arabic numerals can dissociate from other words.                                                            |
| [103] | 2007 | Etchepareborda et al.       | Review<br>—                   | Special skills (hypermnnesia, hyperlexia, hypercalculia, calendar calculation, music, art) in Asperger syndrome as levers for employment.                                                                          |
| [73]  | 2007 | Newman et al.               | Group study<br>3 groups       | ASD+HPL beat ASD–HPL on word and pseudoword decoding and matched reading-age-matched typical readers on everything except comprehension; same basic model of word reading.                                         |
| [119] | 2008 | Cardoso-Martins & Silva †   | Case / series<br>1            | Brazilian case study of how an autistic hyperlexic child learned to read (bibliographic record only).                                                                                                              |
| [72]  | 2009 | Saldaña et al.              | Group study<br>14 + 12        | Adolescents with ASD and a reading–comprehension gap: outstanding readers (vs verbal IQ) had stronger lexical orthographic and phonological representations but no sublexical, rapid-naming or memory advantage.   |
| [43]  | 2009 | Suzuki et al.               | Acquired (adult)<br>1         | Acquired hyperlexia with ambient echolalia after left anterior cingulate infarct; faulty inhibition of intact subroutines.                                                                                         |
| [67]  | 2010 | Castles et al.              | Case / series<br>2            | Two cases read irregular words normally even when they could not define them: evidence for a direct lexical route from print to sound not mediated by meaning.                                                     |
| [110] | 2010 | Joshi et al.                | Case / series<br>2            | Kannada–English bilinguals: the hyperlexic case decoded well but comprehended poorly in <b>both</b> languages; profiles cut across language boundaries.                                                            |
| [34]  | 2011 | Treffert                    | Theory / commentary<br>—      | Hyperlexia, "Einstein syndrome" (late talkers) and "blindisms" can mimic autism; careful differentiation matters for treatment and prognosis.                                                                      |
| [74]  | 2012 | Samson et al.               | Meta-analysis<br>fMRI studies | Autistic participants showed more temporal, occipital and parietal activity and less frontal activity across faces, objects and words: enhanced perceptual resource allocation that may underlie hyperlexia.       |
| [120] | 2013 | Cardoso-Martins et al.<br>† | Case / series<br>1            | Used a 4-year-old's invented spellings to probe the mechanisms of hyperlexic word reading (bibliographic record only).                                                                                             |
| [66]  | 2013 | Lamônica et al.             | Case / series<br>6            | Six Brazilian boys (4;4–5;2), hyperlexia noticed before 36 months: all recognised letters and numbers; most did not understand texts they read.                                                                    |
| [86]  | 2013 | Lin et al.                  | Group study<br>300 + 35       | Six-subtest online assessment (decoding, homographs, auditory/visual vocabulary and sentence comprehension) to identify hyperlexic learning profiles; differentiated autistic children with up to 92% accuracy.    |
| [22]  | 2013 | Mottron et al.              | Theory / commentary<br>—      | Veridical mapping: perception couples with isomorphic structures (letters ↔ sounds, pitch ↔ note names), explaining the shared structure of hyperlexia, absolute pitch and synaesthesia.                           |
| [57]  | 2014 | Pacheva et al.              | Case / series<br>1            | 13-year-old with TSC who did not speak yet read, wrote and comprehended well, with hyperlexia, hypermnnesia and hypercalculia.                                                                                     |
| [36]  | 2014 | Treffert                    | Review<br>—                   | Up to 1 in 10 autistic people have savant abilities; ~50% of savant cases involve autism; IQ can be superior; skills grow from duplication to improvisation to creation.                                           |
| [59]  | 2015 | Stein et al.                | Case / series<br>1            | Gifted (verbal IQ 143), hyperlexic boy improved markedly with Floortime and social/pragmatic group speech therapy; raises the "too bright for peers" question.                                                     |
| [49]  | 2015 | Wei et al.                  | Group study<br>national       | Nationally representative US sample (6–9 y): profiles higher-achieving 39%, hyperlexia 9%, hypercalculia 20%, lower-achieving 32%; all lost ground in passage comprehension over time.                             |
| [18]  | 2017 | Ostrolenk et al.            | Review<br>82 cases; 912       | Systematic review of 82 cases and group studies (912 participants, 315 hyperlexic): 6–21% of autistic children; 84% of cases autistic; chronologically inverted path to reading using the visual word-form system. |
| [31]  | 2018 | Sparks & Luebbers           | Group study<br>random sample  | Most US high-school Spanish learners fit the hyperlexic profile (decoding > comprehension) in their second language: "reading without meaning" is normal in early L2 learning.                                     |
| [54]  | 2018 | Tobia et al.                | Group study<br>18             | In 22q11.2 deletion syndrome, more than two-thirds showed reading above IQ expectations ("hyperlexia"); literacy kept pace with development.                                                                       |
| [53]  | 2019 | Åsberg et al.               | Group study<br>10 + 26        | Among 36 girls with ADHD, 10 had a hyperlexic-like style; these girls had more social-communication difficulties (ADOS-G, ADI-R) but similar IQ and vocabulary.                                                    |
| [50]  | 2019 | Åsberg et al.               | Group study<br>53             | Population cohort at age 8: 25 poor readers, 18 skilled readers, 10 "hyperlexic/poor comprehenders". Oral-language weaknesses at age 3 predicted the hyperlexic profile five years later.                          |
| [65]  | 2019 | Rendall et al.              | Animal model<br>mice          | Autism-risk gene model (SHANK3) shows enhanced low-level pitch discrimination; frames language profiles "from nonverbal to hyperlexic".                                                                            |
| [23]  | 2021 | Macdonald et al.            | Group study<br>3 groups       | ASD+HPL had advanced word reading and letter naming without matching phonological awareness, letter–sound knowledge or language: a non-phonological route to early reading.                                        |
| [48]  | 2021 | Solazzo et al.              | Group study<br>155            | 9% of preschoolers with ASD showed early hyperlexic traits; associated with more repetitive behaviours but also more social-oriented behaviour and better expressive and written communication one year later.     |

| Ref                                                               | Year | Source                   | Design · n                     | Unique contribution                                                                                                                                                                                                                       |
|-------------------------------------------------------------------|------|--------------------------|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [24]                                                              | 2022 | Macdonald et al.         | Group study<br>30              | Parent-supported tablet program (word-, phrase-, sentence-level comprehension; word–picture matching): significant reading comprehension gains for ASD+HPL vs TD ( $p = .023$ ); receptive language gains in all groups.                  |
| [41]                                                              | 2022 | Mammarella et al.        | Case / series<br>2             | Two autistic early readers with above-average IQ and text comprehension (one weak only in oral text) challenge the comprehension-deficit definition.                                                                                      |
| [71]                                                              | 2023 | Luo & Su                 | Case / series<br>1 + controls  | Chinese hyperlexic reading without advanced metalinguistic skills (weak morphology): likely direct whole-character-to-sound mapping.                                                                                                      |
| [20]                                                              | 2023 | Ostrolenk et al.         | Case / series<br>2             | Montréal twins (4–8 y) whose language was only letters and numbers until age 4 then broadened to full sentences: hyperlexic skills can be harnessed toward oral language.                                                                 |
| [21]                                                              | 2024 | Ostrolenk et al.         | Group study<br>701; 313        | Montréal clinic cohort: 22–37% of autistic children had intense/exclusive interest in letters (OR 2.78) and numbers (OR 3.49) despite 76% being minimally verbal; interest emerged ~30 months, independent of oral language.              |
| [79]                                                              | 2025 | Rosselló et al.          | Case / series<br>1             | Minimally verbal 11-year-old with trilingual hyperlexia (Spanish, Catalan, English) and a print-acquired lexicon processed visual recursion like typical peers.                                                                           |
| <b>A2 · Related literature (targeted searches and frameworks)</b> |      |                          |                                |                                                                                                                                                                                                                                           |
| [25]                                                              | 1986 | Gough & Tunmer           | Theory / commentary<br>—       | The Simple View of Reading: reading comprehension = decoding × language comprehension.                                                                                                                                                    |
| [26]                                                              | 2006 | Mottron et al.           | Theory / commentary<br>—       | Enhanced Perceptual Functioning model: perception plays a larger, more autonomous role in autistic cognition.                                                                                                                             |
| [106]                                                             | 2008 | Brenton et al.           | Case / series<br>1             | One of the youngest reported cases of absolute pitch in autism; may become a lifelong skill.                                                                                                                                              |
| [107]                                                             | 2008 | Heaton et al.            | Case / series<br>1 + controls  | Autistic adult with absolute pitch named the pitch of speech far better than controls with AP.                                                                                                                                            |
| [83]                                                              | 2013 | Brown et al.             | Meta-analysis<br>36 studies    | Comprehension gap $g = -0.7$ SD; semantic knowledge explained 57% and decoding 55% of variance; highly social texts were harder; autism alone does not predict comprehension difficulty.                                                  |
| [75]                                                              | 2014 | Bouvet et al.            | Case / series<br>1             | Case "FC" combines savant skills, absolute pitch and synaesthesia-like associations, supporting a common veridical-mapping mechanism.                                                                                                     |
| [98]                                                              | 2014 | El et al.                | Review<br>12 studies           | Effective components: question generation, graphic organisers, predictions, anaphoric cueing, explicit instruction and grouping.                                                                                                          |
| [96]                                                              | 2014 | Ng & Chia                | Case / series<br>small n       | Scaffolding Interrogative Method (SIM): comprehension improved in every intervention phase; comprehension age rose without reading-age change, closing the hyperlexic gap (secondary sources report +19.4% to +36.3% in 3 months).        |
| [109]                                                             | 2016 | Stephenson et al.        | Group study<br>8–11 & 16–18 y  | Reduced skin-conductance response to music-evoked emotion in ASD; age × diagnosis interaction for identifying scary music.                                                                                                                |
| [111]                                                             | 2018 | Gonzalez-Barrero & Nadig | Group study<br>30 + 47         | Montréal study: current language exposure explained 62% of vocabulary and 49% of morphology variance in both autistic and typical children; many autistic children acquire two languages.                                                 |
| [87]                                                              | 2018 | Loukusa et al.           | Group study<br>16 + 16         | Autistic children struggled most with context-dependent inferences, more so when mind-reading was needed, and with explaining how they used context.                                                                                      |
| [112]                                                             | 2019 | Gonzalez-Barrero & Nadig | Group study<br>13 + 13         | Bilingual autistic children scored in the average monolingual range for receptive vocabulary; bilingual exposure is not detrimental.                                                                                                      |
| [104]                                                             | 2019 | Quintin                  | Review<br>—                    | Music (pitch perception, musical memory, emotion identification) is a relative strength in ASD and a strength-based route to social and communication goals.                                                                              |
| [19]                                                              | 2019 | Zhang & Joshi            | Meta-analysis<br>meta-analysis | Supports the profile of good decoding with poor listening and reading comprehension; 66.22% of identified hyperlexics showed precocious decoding; ~96% of studies drew on atypical populations; decoding definitions drive heterogeneity. |
| [90]                                                              | 2020 | McIntyre et al.          | Group study<br>81              | 81 autistic youth aged 8–16, three waves 15 months apart: narrative and inference growth relate to reading comprehension outcomes.                                                                                                        |
| [91]                                                              | 2021 | Ober et al.              | Group study<br>45 + 43         | In autistic adolescents (mean 14.9 y) task-switching predicted comprehension indirectly through decoding.                                                                                                                                 |
| [105]                                                             | 2021 | Romani et al.            | Review<br>17 articles          | Systematic review: absolute pitch prevalence in autism 5–11% with enhanced pitch discrimination; suggests using it for joint attention and communication.                                                                                 |
| [92]                                                              | 2021 | Solis et al.             | Case / series<br>5             | Middle-school vocabulary and main-idea intervention with text choice; upward stable trends; students valued choosing their texts.                                                                                                         |
| [84]                                                              | 2021 | Sorenson et al.          | Meta-analysis<br>26 studies    | In autistic 6–18-year-olds both word reading (mean $r = .65$ ) and oral language ( $r = .61$ ) correlate similarly with comprehension.                                                                                                    |
| [100]                                                             | 2022 | Geretsegger et al.       | Meta-analysis<br>26 RCTs; 1165 | Moderate-certainty benefits for global improvement (RR 1.22), quality of life (SMD 0.28) and symptom severity (SMD $-0.83$ ); unclear effects on social interaction and communication.                                                    |
| [95]                                                              | 2022 | Madaus et al.            | Group study<br>10              | Parents of 10 twice-exceptional autistic college students described best and hardest aspects of high school and transition supports.                                                                                                      |

| Ref   | Year | Source            | Design · n                       | Unique contribution                                                                                                                                                  |
|-------|------|-------------------|----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [88]  | 2022 | Martelle & Namazi | Review<br>—                      | Idiom comprehension depends on inferencing; transparency, familiarity and context help; explicit teaching recommended.                                               |
| [108] | 2023 | Sivathasan et al. | Group study<br>25 + 23           | Autistic children (6–13 y) showed a relative strength recognising emotion in music, with typical accuracy for faces and voices.                                      |
| [99]  | 2025 | Lee et al.        | Meta-analysis<br>5 SCED          | Pictorial/graphic representations produced moderate-to-strong comprehension gains (Tau-U = 0.85); paper-based more consistent, technology-based strong but variable. |
| [94]  | 2025 | Reis & Renzulli   | Review<br>—                      | Synthesis of a Javits-funded programme: strength- and interest-based teaching supports social-emotional health and achievement of 2e autistic students.              |
| [89]  | 2026 | Cheng et al.      | Meta-analysis<br>6 fMRI; 95 + 98 | Shared figurative-language network (bilateral STG, right insula) but consistent hypoactivation of left STG/MTG in autism.                                            |

### A3 · Clinical and educational sources

|       |      |                                     |                 |                                                                                                                                                                                  |
|-------|------|-------------------------------------|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [32]  | 2023 | Whiteside M                         | Web source<br>— | Argues hyperlexia is a genuine reading disorder under the Simple View; recommends IQ, language, comprehension and autism evaluation.                                             |
| [101] | 2025 | And Next Comes L                    | Web source<br>— | Seven families of visual supports (school, comprehension, communication, regulation, executive function, social, self-advocacy): "include the written word as much as possible". |
| [27]  | n.d. | Cleveland Clinic                    | Web source<br>— | Clinical overview: signs (books over toys, spelling aloud or in the air), no single test, multidisciplinary evaluation, speech/occupational/psychological therapy.               |
| [97]  | n.d. | McGill University Newsroom          | Web source<br>— | Lay summary of the McGill tablet intervention; comprehension work can start as early as age 3; app planned as open source.                                                       |
| [30]  | n.d. | Meaningful Speech                   | Web source<br>— | Many hyperlexic children are gestalt language processors; model language in writing, keep letters available, keyboard AAC, turn on subtitles.                                    |
| [37]  | n.d. | The Transmitter (formerly Spectrum) | Web source<br>— | Hyperlexia III case ("Garret") who later made his college Dean's list; Treffert collected 200+ parent reports.                                                                   |
| [28]  | n.d. | SSM Health Treffert Center          | Web source<br>— | Three types; early readers who speak late are often misdiagnosed; speech, occupational and play-based therapy; use written language to teach.                                    |
| [102] | n.d. | Twinkl                              | Web source<br>— | Interest in systems (periodic table, dates, flags, languages); leverage rote learning, special interests, written language ("when in doubt, write it out"), and task breakdown.  |

## APPENDIX B · GLOSSARY

|                                        |                                                                                                  |
|----------------------------------------|--------------------------------------------------------------------------------------------------|
| <b>Decoding</b>                        | Turning print into pronunciation (reading aloud), with or without understanding.                 |
| <b>Orthographic processing</b>         | Recognising written word forms and spelling patterns as visual units.                            |
| <b>Phonological awareness</b>          | Awareness of the sound units of speech (syllables, phonemes).                                    |
| <b>Simple View of Reading</b>          | Reading comprehension = decoding × language comprehension.                                       |
| <b>Pragmatics</b>                      | Using language in social context: tone, turn-taking, implied meaning.                            |
| <b>Gestalt language processing</b>     | Acquiring language in whole chunks (scripts) before analysing single words.                      |
| <b>Enhanced Perceptual Functioning</b> | Model in which perception has a stronger, more autonomous role in autistic cognition.            |
| <b>Veridical mapping</b>               | Perceptual detection of structural correspondences (e.g., letters ↔ sounds, pitch ↔ note names). |
| <b>Hypercalculia / hypernumeracy</b>   | Precocious fascination with and skill in numbers.                                                |
| <b>Absolute pitch</b>                  | Naming a musical note without a reference tone.                                                  |
| <b>Short vs long scale</b>             | Large-number naming systems: English billion = $10^9$ ; French billion = $10^{12}$ .             |
| <b>Twice-exceptional (2e)</b>          | Gifted and also neurodivergent or disabled.                                                      |
| <b>OR / HR / RR</b>                    | Odds, hazard and risk ratios; 1 means no difference.                                             |
| <b>SMD / g</b>                         | Standardised mean difference; about 0.2 small, 0.5 medium, 0.8 large.                            |
| <b>Tau-U</b>                           | Single-case effect size combining trend and non-overlap; 0.85 is large.                          |
| <b>SCED</b>                            | Single-case experimental design.                                                                 |
| <b>ALE</b>                             | Activation likelihood estimation, a neuroimaging meta-analysis method.                           |